

Step	Reason
1. $(p \wedge t) \Rightarrow (r \vee s)$	Premise
2. $q \Rightarrow (u \wedge t)$	Premise
3. $u \Rightarrow r$	Premise
4. $\neg s$	Premise
5. $\neg(p \wedge t) \vee (r \vee s)$	Conditional-disjunction equivalence on 1
6. $\neg p \vee \neg t \vee r \vee s$	1st De Morgan's law on 5
7. $s \vee \neg p \vee \neg t \vee r$	Commutative law on 6
8. $\neg p \vee \neg t \vee r$	Disjunctive syllogism on 7, 4
9. $\neg t \vee r$	Conditional-disjunction equivalence on 3
10. $p \vee \neg t$	Commutative law on 9
11. $\neg t \vee \neg r \vee r$	Resolution principle on 10, 8
12. $\neg(t \wedge r)$	1st De Morgan's Law on 11
13. $(u \wedge t) \Rightarrow r$	Conditional-disjunction equivalence on 12
14. $q \Rightarrow r$	Hypothetical syllogism on 2, 13

13) For each of these arguments, explain which rules of inference are used for each step

a) "Doug, a student in this class, knows how to write programs in JAVA. Everyone who knows how to write programs in JAVA can get a high-paying job. Therefore, someone in this class can get a high-paying job."

Soln: $S(x)$: x is a student of this class $J(x)$: x knows how to write programs in Java $Job(x)$: x can get a high-paying job.

Premise: 1) $S(\text{Doug}) \wedge J(\text{Doug})$ 2) $\forall x (J(x) \Rightarrow Job(x))$

Conclusion: $\exists x (S(x) \wedge Job(x))$

Domain: Set of all people in the world.